LIFE CYCLE PHASE 1: CONCEPT PHASE

PROJECT CHARTER
Automotive Airbag ECU Functional Safety Project

Document Ref: LCP01-AIRBAG-ECU-001

Date: June 19, 2026

Classification: Safety-Critical (ASIL D / SIL 3)

Project Manager: Alita Shrestha

Institution: Hochschule Harz (HS Harz)

Version: 1.00

PROJECT TITLE

Development and Functional Safety Compliance Certification of an Automotive Airbag Electronic Control Unit (ECU) Simulation System

PROJECT DESCRIPTION

This project encompasses the end-to-end design, implementation, verification, and validation of a safety-critical automotive Airbag Electronic Control Unit (ECU) prototype. Operating in a dual-channel **2oo2 (Two-out-of-Two) redundant hardware and software architecture**, the system utilizes twin identical Arduino Due processing nodes built upon the SAM3X8E ARM Cortex-M3 microcontroller running independent crash-detection and sensor-fusion algorithms.

The physical system reads data from four simulated safety sensors: an accelerometer, a gyroscope, a pressure sensor (realized via localized 10kΩ potentiometer sensor-simulation circuits), and a digital switch representing an Occupant Classification System (OCS) for baby seat detection. Inter-ECU status validation and diagnostic heartbeats are exchanged via a localized high-speed Controller Area Network (CAN Bus) at 500 kbit/s. False deployment prevention is physically guaranteed at the hardware layer via a specialized quad 2-input DM74LS08N AND-gate framework acting as the safety voting core. The entire project lifecycle is structurally organized and executed under a strict Functional Safety Management (FSM) framework to maintain full compliance with international standards **ISO 26262:2018** and **IEC 61508:2010**.

CRITICAL INTEGRITY TARGET: System-level classification is designated at ISO 26262 ASIL D and IEC 61508 SIL 3. Target Probabilistic Metric for random Hardware Failures (PMHF) is strictly constrained to < 10^{-8}/h.

PROJECT PURPOSE

In modern automotive engineering, the Airbag ECU is a supreme safety-critical component. Any functional failure represents an immediate life-threatening risk: a systemic or random failure could result in non-deployment during a high-energy collision event, or conversely, an inadvertent deployment under normal highway driving conditions.

The fundamental purpose of this project is to apply a rigorous structural lifecycle framework that systematically neutralizes human design errors (via quality management) and detects/controls random hardware structural failures. By engineering a dual-layer hardware/software 2oo2 voting topology, the project demonstrates how a highly dependable fault-tolerant architecture can achieve full safe-state management (inhibiting pyrotechnic triggering during hardware anomalies or communication timeouts), thereby proving compliance with strict global automotive type-approval safety standards.

PROJECT OBJECTIVES

- **Architecture Implementation:** Establish a fully operational dual-channel hardware-in-the-loop (HIL) prototype utilizing twin SAM3X8E microcontrollers executing peer-equal monitoring loops.
- **Algorithmic Crash Detection:** Develop an optimized sensor fusion algorithm achieving a real-time crash-detection latency of **≤ 15 ms** upon exceeding the defined multi-sensor deployment thresholds.
- **Hardware-Enforced Voting:** Implement a fail-safe voting mechanism where deployment outputs are physically blocked by an autonomous hardware AND-gate matrix unless both microcontrollers independently issue a synchronized HIGH fire signal.
- **High Integrity Code Coverage:** Attain **100% Statement Coverage, 100% Branch Coverage, and 100% Modified Condition/Decision Coverage (MC/DC)** for all ASIL D deployment logic conditions (specifically the `voteDeploy` truth matrix).
- **Traceable Safety Case Lifecycle:** Compile a comprehensive engineering dossier across the 16 lifecycle phases governed by IEC 61508, ensuring precise bi-directional traceability from Hazard Analysis up to final Integration Test Reports.

SCOPE STATEMENT

The scope of this project covers the lifecycle development phases of the Airbag ECU simulation system from the initial conceptualization, risk mapping, and architectural layout, through software block coding, to laboratory hardware synthesis and rigorous fault-injection testing. The development boundaries are established by the simulated hardware environment provided by the Department of Automation and Computer Science at HS Harz, utilizing physical microcontrollers to execute real safety routines, with pyrotechnic loops simulated via fast-acting visual diagnostic LEDs.

In-Scope

- Conducting a full **Hazard Analysis and Risk Assessment (HARA)** to isolate 6 distinct hazardous events and formulate 7 target Safety Goals (SG-01 to SG-07).
- Drafting 22 Functional Safety Requirements (FSRs) and 22 Software/Hardware Safety Requirements (SRs).
- Full implementation of 6 distinct software modules (MOD-01 to MOD-06) covering Power-On Self-Test (POST), Acquisition, Crash Detection, Deployment

Out-of-Scope

- Development of mass-production automotive grade multilayer PCBs or application-specific integrated circuits (ASICs).
- Physical integration or live firing of actual explosive pyrotechnic squibs, gas generators, or physical canvas fabric airbag cushions.
- Full-scale environmental stress testing within specialized thermal or electromagnetic compatibility (EMC) chambers (handled theoretically via LV124/ISO16750-3 design planning).

Control, CAN Communication, and Diagnostic Monitoring.

- Configuring 500 kbit/s Classical CAN inter-node bus topology utilizing Waveshare SN65HVD230 transceivers with localized 15-bit hardware CRC checking.
- Engineering occupant classification logic supporting a 10ms software debounce filter for baby seat suppression on Airbag 2.
- Execution and document tracking of 16 comprehensive Integration Test cases (IT-01 to IT-16), documenting normal states, edge boundaries, and aggressive fault-injections.

- Integration of SOTIF (ISO 21448) or Cybersecurity (ISO 21434) controls beyond basic defensive communication layers.
- Interfacing with real commercial automotive chassis backbone networks (e.g., FlexRay, Automotive Ethernet) outside of the dedicated inter-ECU diagnostic CAN channel.

HIGH-LEVEL REQUIREMENTS

The system must satisfy the following technical and functional safety constraints to achieve validation:

System Performance & Safety Co-Constraints:

- **Single-Point Fault Metric (SPFM):** Diagnostic safety mechanisms must provide a minimum of **≥ 99%** single-point fault coverage for all ASIL D components.
- **Execution Loop Latency:** The main background execution loop on both Arduino nodes must consistently cycle within ≤ **1 ms**.
- **Fault-to-Safe Latency (DWI):** Activation of the shared Diagnostic Warning Indicator (DWI) fault lamp must occur within ≤ **1 ms** of any local fault or within ≤ **800 ms** during complex multi-node system diagnostic evaluation.
- **Communication Timeout Safe-State:** If valid peer CAN data frames are absent for more than **2000 ms**, the system must trigger a Communication Fault, transition to a safe state, and completely inhibit deployment to prevent erratic firing (documented timing deviation).
- **POST Pre-Conditions:** Hardware outputs must remain strictly pulled to a default LOW state until the completion of Power-On Self-Tests, which must run a complete CPU multiply validation, a 128-word RAM checkerboard structural evaluation, and an analog ADC range test.

DELIVERABLES

The project execution requires the compilation and submission of the following discrete engineering artifacts:

DELIVERABLE ID & TITLE	LIFECYCLE PHASE	TECHNICAL CONTENT & DESCRIPTION
D01: Lifecycle Engineering Dossier	LCP01 - LCP05	Comprehensive safety dossier including Item Definition, Hazard Analysis (HARA), Functional Safety Requirements Specification (FSRS v1.0), and the System Architecture Safety Concept Document (SCD).
D02: 2oo2 Hardware Prototype	LCP09 (Implementation)	Fully assembled Hardware-in-the-Loop (HIL) lab bench configuration incorporating dual SAM3X8E Arduino nodes, Waveshare CAN transceivers, DM74LS08N logic gate ICs, potentiometer array, and LED diagnostic monitors.
D03: ASIL D Compliant Firmware	LCP07 - LCP08	Modular C/C++ embedded repository containing verified source code for MOD-01 through MOD-06, featuring hardware watchdog timer (WDT) configurations and bit-mapped status CAN registries.
D04: V&V Framework & Reports	LCP13 (Testing)	The complete test bench execution plan tracing 16 integration tests (IT-01 to IT-16), containing specific serial telemetry output capture logs, alongside formal QA Reports v1.0 and v2.0 confirming 100% MC/DC alignment.
D05: Technical Presentation	Project Evaluation	The final technical compliance presentation delivered to the Department of Automation and Computer Science demonstrating a complete, validated system safety case.

PROJECT ORGANIZATION & RESPONSIBILITIES

To enforce professional segregation of duties required by functional safety standards, the following roles are formally established for the 14-member project group:

FUNCTIONAL SAFETY ROLE	ASSIGNED TEAM MEMBER	CORE GOVERNANCE DUTY
Project Manager	Alita Shrestha	Project chartering, schedule tracking, milestone approvals.
Functional Safety Manager	Kruthika Sathyanarayana Gowda	Safety plan oversight, standard compliance, sign-off authority.
Safety Requirement Engineers	Fateme Hosseinzadeh, Mobina Ketabi	Elicitation and bi-directional traceability of FSRs and SWRs.
Software Architect & Engineer	Mariana Chavez Avila, Mrunal Sahebrao Ghotekar	Modular software architectural design and embedded execution loop coding.
Quality Manager	Thirthashri Gowli Ganesh	Process audits, configuration control, QA Report v1.0/v2.0 validation.
Safety Engineers	Paresh Sanjiv Patil, Atharva Banarase	Safety mechanism specification, diagnostic coverage optimization.
Safety Analysts (HARA)	Sara Nematollahi, Sathyashree Vignasante Ramegowda	Hazard assessment, risk parameter mapping, and ASIL determination.
System Engineer & Architect	Atharva Prasad Avsare, Om Naik	2oo2 hardware block schematics and BOM procurement.
Testing Engineer V&V	Shivbaraje Bhosale	Integration test bench construction, test scripting, and bug logging.